JNTUH USED ON 15-05-18 AM Set 1

Code No: 117EE **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, April/May - 2018
LINUX PROGRAMMING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a)	Write a short note on Filters.	[2]
b)	What is a Field Buffer and a Record Buffer.	[3]
c)	Explain briefly FIFO file in UNIX File System.	[2]
d)	Write a short note on "mkdir" command.	[3]
e)	Write a short note on Signal function.	[2]
f)	Compare and Contrast Reliable and Unreliable Signals.	[3]
g)	Write a short note on UNIX system V IPC methods.	[2]
h)	Write a short note on "rclose()" library function.	[3]
i)	Explain briefly about "setsockopt" system call.	[2]
j)	Write a short note on Berkeley Socket.	[3]

PART-B

(50 Marks)

2. Explain in detail about Control Structures. [10]

OR

3.a) Explain in detail about Process Utilities.
b) Write a short notes on security using file permissions. [5+5]

4. Explain in detail about following File Operations:
a) seek b) fcntl [5+5]

OR

5. Explain in detail about File and Record Locking in UNIX. [10]

6. Explain in detail about Kernel support for Signals. [10]

OR

7. Write a detailed note on following processes:
a) Zombie processes b) Orphan processes. [5+5]

8. Explain in detail about IPC between related processes using unnamed pipes. [10]

OR

9. Explain in detail about Kernel support for Semaphores. [10]

10. Make a comparison of various IPC mechanisms. [10]

OR

11. Explain the usage of Stream Sockets using Client-Server Message handling example. [10]

--ooOoo--

JNTUH USED 15-05-18AM Set